

1. MC.03.01.ALGO (Algorithmic)

A factor that causes or leads to a change in a cost or activity is a(n)

-  ☒ a. driver.
- ☐ b. expense.
- ☐ c. cost object.
- ☐ d. allocation.
- ☐ e. transaction.

Feedback

Check My Work

A factor measuring the output of an activity resulting in a change in costs. Look at what causes the activity, the output measure.

Solution
a

2. MC.03.02.ALGO (Algorithmic)

Which of the following would probably be a variable cost in a soda bottling plant?

-  ☒ e. All of these
- ☐ a. Carbonated water
- ☐ b. Bottle caps
- ☐ c. Direct materials
- ☐ d. Cost of the labels printed on the bottles

Feedback

Check My Work

Variable costs move directly with production activity.

Solution
e

3. MC.03.04.ALGO (Algorithmic)

The following cost formula was developed by using monthly data for a hospital.

Total Cost = \$128,000,000 + (\$12,000 × Number of Patient Days)

In the cost formula, the term \$128,000,000

-  ☒ c. is the total fixed cost.
- ☐ a. is the dependent variable.
- ☐ b. is the cost driver.
- ☐ d. is the variable rate.
- ☐ e. None of these choices are correct.

Feedback

Check My Work

Review cost equation. The total cost formula consists of fixed costs (constant) and variable costs (variable rate x number of units). Fixed costs correspond with the y-intercept. Variable costs reflect the slope. Dependent variable is the cost being predicted. Independent variable is the cost driver.

Solution

c

4. MC.03.05.ALGO (Algorithmic)

The following cost formula was developed by using monthly data for a hospital.

$$\text{Total Cost} = \$128,000,000 + (\$12,000 \times \text{Number of Patient Days})$$

In the cost formula, the term \$12,000

- ✓
- ☐ a. is the predetermined manufacturing overhead allocation rate.
 - ☐ b. is the intercept.
 - ✓ ☒ c. is the variable rate.
 - ☐ d. is the fixed cost driver.
 - ☐ e. None of these choices are correct.

Feedback

Check My Work

Review cost equation. The total cost formula consists of fixed costs (constant) and variable costs (variable rate x number of units). Fixed costs correspond with the y-intercept. Variable costs reflect the slope. Dependent variable is the cost being predicted. Independent variable is the cost driver.

Solution

c

5. MC.03.08

The following cost formula for total purchasing cost in a factory was developed using monthly data.

$$\text{Total Cost} = \$235,000 + (\$75 \times \text{Number of Purchase Orders})$$

Next month, 8,000 purchase orders are predicted. The total cost predicted for the purchasing department next month

- ✓
- ☐ a. is \$8,000.
 - ☐ b. is \$235,000.
 - ☐ c. is \$600,000.
 - ✓ ☒ d. is \$835,000.
 - ☐ e. cannot be determined from the above formula.

Feedback

Check My Work

Calculate the total cost using the cost formula and the number of purchase orders given.

Post-Submission

$$\text{Total Cost} = \$235,000 + (\$75 \times 8,000) = \$835,000$$

Solution

d

6. MC.03.10

An advantage of the high-low method is that it



- ☐ a. is subjective.
- ☒ b. is objective.
- ☐ c. is the most accurate method.
- ☐ d. removes outliers.
- ☐ e. is descriptive of nonlinear data.

Feedback

Check My Work

Review high-low method. This method separates mixed costs. High and low points are determined by activity.

Solution
b

7. MC.03.12

The following 6 months of data were collected on maintenance cost and the number of machine hours in a factory:

Month	Maintenance Cost	Machine Hours
January	\$16,900	5,600
February	13,900	4,500
March	10,900	3,800
April	11,450	3,700
May	13,050	4,215
June	16,990	4,980

Select the correct set of high and low months.



- ☒ a. High: January, Low: April
- ☐ b. High: January, Low: March
- ☐ c. High: June, Low: March
- ☐ d. High: June, Low: April

Feedback

Check My Work

High and low points are determined by activity.

Solution
a

8. MC.03.14

The total cost for monthly supervisory cost in a factory is \$4,500 regardless of how many hours the supervisor works or the quantity of output achieved. This cost



- ☐ a. is strictly variable.
- ☒ b. is strictly fixed.

- ☐ c. is a mixed cost.
- ☐ d. is a step cost.
- ☐ e. cannot be determined from this information.

Feedback

Check My Work

Variable costs move directly with production activity. Fixed costs remain constant with no influence by the production activity within the relevant range. These costs are part of the business regardless of sales or production amounts.

Solution
b

9. MC.03.16

Yates Company shows the following unit costs for its product:

Direct materials	\$40
Direct labor	30
Variable overhead	2
Fixed overhead	5

Yates started the year with 8,000 units in inventory, produced 50,000 units during the year, and sold 55,000 units. The value of ending inventory is

- ☐ a. greater under variable costing than absorption costing.
- ☐ b. This situation cannot happen.
- ☐ c. the same under both variable and absorption costing.
- ☐ d. There is no ending inventory.
- ☒ e. greater under absorption costing than variable costing.

Feedback

Check My Work

The difference between variable and absorption costing is the treatment of fixed factory overhead. Absorption costing includes fixed factory overhead in the cost of the product.

Solution
e

10. MC.03.17

In the method of least squares, the coefficient that tells the percentage of variation in the dependent variable that is explained by the independent variable is

- ☐ a. the intercept term.
- ☐ b. the x -coefficient.
- ☐ c. the coefficient of correlation.
- ☒ d. the coefficient of determination.
- ☐ e. none of these.

Check My Work

The measure of activity output.

Solution
d

11. PR.03.237.ALGO (Algorithmic)

The controller at On Time Delivery wants to break out the cost of deliveries into fixed and variable components so that the company will be better able to predict costs for next year. The delivery costs incurred each month and the number of deliveries are as follows:

<u>Month</u>	<u>Delivery Cost</u>	<u>Deliveries</u>
January	\$2,190.00	106
February	2,250.00	115
March	1,948.60	90
April	1,700.00	81
May	1,645.00	76
June	2,760.00	138
July	3,675.00	181
August	1,935.20	93
September	1,739.60	78
October	3,840.00	181
November	3,900.00	199
December	4,065.00	201

Using the high-low method:

A. Calculate the variable cost rate per deliveries. **(Round answer to two decimal places.)**

\$ 

19.36

B. Calculate the fixed cost of overhead. **(Round answer to two decimal places.)**

\$ 

173.64

C. Construct the cost formula for total overhead cost. **If required, round your answer to two decimal places.**

$$\begin{aligned}
 \text{Total Cost} &= \text{Fixed Cost} + \text{Variable Cost} \\
 &= \$ \text{  } 173.64 + (\$ \text{  } 19.36 \times \text{Number of Deliveries})
 \end{aligned}$$

D. The company is estimating that in January the number of deliveries will be 113. How much should it estimate to have in total delivery costs for January? **(Round answer to two decimal places.)**

\$ 

2,361.32

Check My Work
Correct
Post-Submission

A. Variable Rate = $(\$4,065 - \$1,645) \div (201 - 76) = \19.36

B. High point:

Fixed Cost = $\$4,065 - (201 \times \$19.36) = \$173.64$

Low point:

Fixed Cost = $\$1,645 - (76 \times \$19.36) = \$173.64$

C. Cost formula:

Total Cost = $\$173.64 + (\$19.36 \times \text{Number of Deliveries})$

D. Total delivery cost for January:

Total Cost = $\$173.64 + (\$19.36 \times 113)$

Total Cost = $\$2,361.32$

Solution

The controller at On Time Delivery wants to break out the cost of deliveries into fixed and variable components so that the company will be better able to predict costs for next year. The delivery costs incurred each month and the number of deliveries are as follows:

<u>Month</u>	<u>Delivery Cost</u>	<u>Deliveries</u>
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November	3,900.00	199
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Using the high-low method:

A. Calculate the variable cost rate per deliveries. **(Round answer to two decimal places.)**

\$

B. Calculate the fixed cost of overhead. **(Round answer to two decimal places.)**

\$

C. Construct the cost formula for total overhead cost. **If required, round your answer to two decimal places.**

$$\text{Total Cost} = \text{Fixed Cost} + \text{Variable Cost}$$

$$= \$ 173.64 + (\$ 19.36 \times \text{Number of Deliveries})$$

D. The company is estimating that in January the number of deliveries will be 113. How much should it estimate to have in total delivery costs for January? **(Round answer to two decimal places.)**

\$ 2,361.32

12. PR.03.229

Consider each of the following independent situations.

For each situation, describe the cost as one of the following: fixed cost, variable cost, mixed cost, or step cost.

- | | | |
|---|-------------------|---|
| A. The salary of a legal secretary in a law firm. | <u>Fixed</u> ▼ | ✓ |
| B. A lease contract for an automobile that requires a monthly payment of \$300 plus \$0.05 per mile. | <u>Mixed</u> ▼ | ✓ |
| C. The cost of lumber for a homebuilder. | <u>Variable</u> ▼ | ✓ |
| D. The cost of Internet service that is calculated based on hours of usage. | <u>Variable</u> ▼ | ✓ |
| E. The cost of telephone service that includes a fixed monthly charge of \$50 plus \$0.10 a minute for long-distance calls. | <u>Mixed</u> ▼ | ✓ |
| F. The salary cost of seasonal tax preparers for a CPA firm; One tax preparer can prepare 100 tax returns per month. | <u>Step</u> ▼ | ✓ |
| G. A factory supervisor's salary. | <u>Fixed</u> ▼ | ✓ |
| H. The cost of sugar in the production of soft drinks. | <u>Variable</u> ▼ | ✓ |

Feedback

Check My Work
Correct
Post-Submission

Variable costs are costs that in total vary in direct proportion to changes in output within the relevant range. For example, for a crayon manufacturing company, the cost of wax and pigments varies with the number of crayons produced. Hence, these costs are variable costs for the crayon manufacturing company.

Fixed costs are costs that in total are constant within the relevant range as the level of output increases or decreases. Fixed costs are extremely important to understand because they usually are relatively large in amount. For example, the rental cost of warehouse space by a wholesaler is fixed for the term of the lease. Whether the wholesaler's sales go up or down, the cost of the leased warehouse stays the same.

Mixed costs are costs that have both a fixed and a variable component. For example, sales representatives often are paid a salary plus a commission on sales.

Step cost is a cost that displays a constant level of total cost for a range of output and then jumps to a higher level of total cost at some point, where it remains for a similar range of output (e.g., renting freezer trucks on an "as-needed" basis to manage baking companies' fixed costs as they temporarily step up during the holiday season due to the increase in their sales volume).

Solution

Consider each of the following independent situations.

For each situation, describe the cost as one of the following: fixed cost, variable cost, mixed cost, or step cost.

- A. The salary of a legal secretary in a law firm. Fixed
- B. A lease contract for an automobile that requires a monthly payment of \$300 plus \$0.05 per mile. Mixed
- C. The cost of lumber for a homebuilder. Variable
- D. The cost of Internet service that is calculated based on hours of usage. Variable
- E. The cost of telephone service that includes a fixed monthly charge of \$50 plus \$0.10 a minute for long-distance calls. Mixed
- F. The salary cost of seasonal tax preparers for a CPA firm; One tax preparer can prepare 100 tax returns per month. Step
- G. A factory supervisor's salary. Fixed
- H. The cost of sugar in the production of soft drinks. Variable

13. PR.03.236.ALGO (Algorithmic)

The overhead costs and labor hours for Adell & Co. are as follows:

<u>Month</u>	<u>Overhead Costs</u>	<u>Labor Hours</u>
January	\$9,670	470
February	12,600	690
March	11,350	560
April	11,830	600
May	13,320	730
June	10,080	520
July	8,260	440
August	6,110	350

Using the high-low method:

- A.** Calculate the variable cost rate per labor hour. **Round your answer to the nearest cent.**

\$ ✓

18.97

- B.** Calculate the fixed cost of overhead. **Round your answer to the nearest cent.**

\$ ✓

-528.10

- C.** Construct the cost formula for total overhead cost. **If required, round your answer to the nearest cent.**

$$\begin{array}{lcl}
 \text{Total Cost} & = & \text{Fixed Cost} + \text{Variable Cost} \\
 & & (\$ \text{ } 18.97 \text{ per Hour} \times \text{Labor Hours}) \\
 & = & \$ -528.10 +
 \end{array}$$

- D.** The company is estimating that in September the labor hours will be 650. How much should it estimate to have in total overhead cost for September? **Round your answer to the nearest cent.**

\$ ✓

11802.40

Check My Work

Correct

Post-Submission

A. Variable Rate = (High Point Cost – Low Point Cost) ÷ (High Point Hours – Low Point Hours)
 = (\$13,320 – \$6,110) ÷ (730 – 350)
 = \$7,210 ÷ 380 = \$18.97/hour variable cost rate

B. High point

Fixed Cost = Total Cost – (Variable Rate × Labor Hours)

\$13,320 – (\$18.97 × 730) = \$-528.10

Low point

Fixed Cost = \$6,110 – (\$18.97 × 350) = \$-529.5

C. Cost formula

Total Cost = \$-529.5 + (\$7,210 × Labor Hours)

D. September = 650 labor hours

Total Cost = \$-528.10 + (\$18.97 × 650)

Total Cost = \$11,802.40

Solution

The overhead costs and labor hours for Adell & Co. are as follows:

<u>Month</u>	<u>Overhead Costs</u>	<u>Labor Hours</u>
January	\$9,670	470
February	12,600	690
March	11,350	560
April	11,830	600
May	13,320	730
June	10,080	520
July	8,260	440
August	6,110	350

Using the high-low method:

A. Calculate the variable cost rate per labor hour. **Round your answer to the nearest cent.**

\$

B. Calculate the fixed cost of overhead. **Round your answer to the nearest cent.**

\$

C. Construct the cost formula for total overhead cost. **If required, round your answer to the nearest cent.**

Total Cost = Fixed Cost + Variable Cost
 = \$ + (\$ per Hour X Labor Hours)

D. The company is estimating that in September the labor hours will be 650. How much should it estimate to have in total overhead cost for September? **Round your answer to the nearest cent.**

\$

14. PR.03.249.ALGO (Algorithmic)

During the most recent year, Boston Corp. had the following data:

Beginning inventory in units	0
Units produced	15,900
Units sold (\$125 per unit)	8,300
Variable costs per unit:	
Direct materials	\$19
Direct labor	12
Variable overhead	18
Fixed costs:	
Fixed overhead per unit produced	\$24
Fixed selling and administrative	168,000

A. How many units are in ending inventory?



7600
units

Feedback

Check My Work

Correct

Post-Submission

A. Ending Inventory in Units = Beginning Units + Units Produced - Units Sold = 0 + 15,900 - 8,300 = 7,600 units

B. Using absorption costing, calculate the per-unit product cost. What is the value of ending inventory?

Per-unit product cost \$

73

Ending inventory \$

554800

Feedback

Check My Work

Correct

Post-Submission

B.

Per unit:

Direct materials	\$19
Direct labor	12
Variable overhead	18

Fixed overhead per unit produced	24
Total product cost	<u>\$73</u>
Ending Inventory Value = $\$73 \times 7,600 = \$554,800$	

C. Using variable costing, calculate the per-unit product cost. What is the value of ending inventory?

Per-unit product cost \$ 

49

Ending inventory \$ 

372400

Feedback

Check My Work
Correct
Post-Submission

C.

Per unit:

Direct materials	\$19
Direct labor	12
Variable overhead	<u>18</u>
Total product cost	<u>\$49</u>
Ending Inventory Value = $\$49 \times 7,600 = \$372,400$	

D. Prepare an income statement using absorption costing.

Boston Corp.
Absorption-Costing Income Statement
For the Most Recent Year

<u>Sales</u>  	\$ 
	1037500
<u>Less: Cost of goods sold</u>  	
	605900
	<u></u>
Gross margin	\$ 
	431600
<u>Less: Selling and administrative expenses</u>  	
	168000
	<u></u>
<u>Operating income</u>  	\$ 
	263600

Feedback

Check My Work
Correct
Post-Submission

D.

Sales (8,300 × \$125) = \$1,037,500

Less cost of goods sold (8,300 × \$73) = \$605,900

E. Prepare an income statement using variable costing.

Boston Corp.
Variable-Costing Income Statement
For the Most Recent Year

<u>Sales</u> ✓	\$ ✓	
	1037500	
<u>Less: Cost of goods sold</u> ✓	✓	
	406700	
	✓	
Gross margin	\$ ✓	
	630800	
Less fixed expenses:		
<u>Overhead</u> ✓	\$ ✓	
	381600	
<u>Selling and administrative expenses</u> ✓	✓	✓
	168000	549600
	✓	
<u>Operating income</u> ✓	\$ ✓	
	81200	

Feedback

Check My Work

Correct

Post-Submission

E.

Sales (8,300 × \$125) = \$1,037,500

Less cost of goods sold (8,300 × \$49) = \$406,700

Overhead (15,900 × \$24) = \$381,600

Feedback

Check My Work

Correct

Solution

1	A. How many units are in ending inventory? 7600 units
2	B. Using absorption costing, calculate the per-unit product cost. What is the value of ending inventory? Per-unit product cost \$ 73

Ending inventory \$

C. Using variable costing, calculate the per-unit product cost. What is the value of ending inventory?

3 Per-unit product cost \$

Ending inventory \$

D. Prepare an income statement using absorption costing.

Boston Corp.
Absorption-Costing Income Statement
For the Most Recent Year

4 Sales	\$	1037500
Less: Cost of goods sold		605900
Gross margin	\$	431600
Less: Selling and administrative expenses		168000
Operating income	\$	263600

E. Prepare an income statement using variable costing.

Boston Corp.
Variable-Costing Income Statement
For the Most Recent Year

Sales		\$ 1037500
5 Less: Cost of goods sold		406700
Gross margin		\$ 630800
Less fixed expenses:		
Overhead	\$ 381600	
Selling and administrative expenses	168000	549600
Operating income		\$ 81200

15. PR.03.248.ALGO (Algorithmic)

Last year, Baker Company produced 35,400 units and sold 25,100 units. Beginning inventory was zero. During the period, the following costs were incurred:

Indirect labor (variable)	\$56,000
Indirect materials (variable)	40,000
Other variable overhead	100,000
Fixed manufacturing overhead	171,000
Fixed administrative expenses	159,000
Fixed selling expenses	105,000
Variable selling expenses per unit	45

Direct labor, per unit	75
Direct materials, per unit	25

Round intermediate calculations to two decimal places and your final answers to the nearest dollar.

A. Compute the dollar amount of ending inventory using the Absorption costing.

\$ 

1136708

B. Compute the dollar amount of ending inventory using the Variable costing.

\$ 

1086959

Feedback

Check My Work
Correct
Post-Submission

A.

Variable costs:

Direct materials	\$25
Direct labor	75
Indirect labor ($\$56,000 \div 35,400$)	1.58
Indirect materials ($\$40,000 \div 35,400$)	1.13
Other variable overhead ($\$100,000 \div 35,400$)	2.82
Variable product costs per unit	\$105.53
Fixed manufacturing overhead ($\$171,000 \div 35,400$)	4.83
Total product costs per unit	\$110.36
Inventory units	$\times 10,300$
Inventory value	<u>\$1,136,708</u>

B.

Variable product costs per unit	\$105.53
Inventory units	$\times 10,300$
Inventory value	<u>\$1,086,959</u>

Solution

Last year, Baker Company produced 35,400 units and sold 25,100 units. Beginning inventory was zero. During the period, the following costs were incurred:

Indirect labor (variable)	\$56,000
Indirect materials (variable)	40,000
Other variable overhead	100,000
Fixed manufacturing overhead	171,000
Fixed administrative expenses	159,000

Fixed selling expenses	105,000
Variable selling expenses per unit	45
Direct labor, per unit	75
Direct materials, per unit	25

Round intermediate calculations to two decimal places and your final answers to the nearest dollar.

A. Compute the dollar amount of ending inventory using the Absorption costing.

\$ 1136708

B. Compute the dollar amount of ending inventory using the Variable costing.

\$ 1086959

16. PR.03.225.ALGO (Algorithmic)

Havier Company manufactures drill presses. Based on past experience, Havier has found that its total overhead cost can be represented by the following formula: Total overhead cost = \$21,000 + \$1.25x, where x = number of machine hours. Last year, Havier incurred 131,000 machine hours.

A. What was the total overhead cost incurred by Havier last year?

\$ ✓

184750

B. What was the total variable overhead cost incurred by Havier last year?

\$ ✓

163750

C. What was the total overhead cost per machine hour last year? **(Round answer to two decimal places.)**

\$ ✓

1.41

D. What was the fixed overhead cost per machine hour last year? **(Round answer to two decimal places.)**

\$ ✓

0.16

E. If Havier incurs 130,500 machine hours next year, what will be the total overhead cost per machine hour? **(Round answer to two decimal places.)**

\$ ✓

1.41

Feedback

Check My Work

Correct

Post-Submission

A. $\$184,750 = \$21,000 + (\$1.25 \times 131,000)$

B. $\$1.25 \times 131,000 = \$163,750$

C. $\$184,750 \div 131,000 = \1.41

D. $\$21,000 \div 131,000 = \0.16

E. $\$184,125 = \$21,000 + (\$1.25 \times 130,500)$

$\$184,125 \div 130,500 = \1.41

Havier Company manufactures drill presses. Based on past experience, Havier has found that its total overhead cost can be represented by the following formula: Total overhead cost = \$21,000 + \$1.25x, where x = number of machine hours. Last year, Havier incurred 131,000 machine hours.

A. What was the total overhead cost incurred by Havier last year?

\$ 184750

B. What was the total variable overhead cost incurred by Havier last year?

\$ 163750

C. What was the total overhead cost per machine hour last year? **(Round answer to two decimal places.)**

\$ 1.41

D. What was the fixed overhead cost per machine hour last year? **(Round answer to two decimal places.)**

\$ 0.16

E. If Havier incurs 130,500 machine hours next year, what will be the total overhead cost per machine hour? **(Round answer to two decimal places.)**

\$ 1.41